

Bring Them Home

Flight Director • Lunar Return Mission

10 SHIFTS • 40 STOPS • GRADE 6 • BRING_THEM_HOME_MS

THE OPENING CARD

Three people are in a spacecraft on the far side of a quarter-million miles, and something aboard has just exploded. You are the flight director, and the calls are yours to make. Power is falling. The cabin is cooling. The trajectory is drifting off the one path that gets them back, and the air they are breathing is already on a countdown. The crew can hear every word said on the loop.

The Failure

BEFORE THE SHIFT — GREET

Four rows of consoles, and every row owns part of the answer

The room is one floor of desks, and the team that reads the power numbers sits two rows from the team that reads the air. Put a name to as many of them as you can before the next pass. In here you ask across the room, not down a corridor.

PLAN CARD 72W

The bang was four minutes ago. Commander Hale heard it before any instrument showed it. Now three consoles disagree. One main power line reads zero. Several cabin pressure readings dropped in the same second. Evelyn Carter says readings that drop together are one fault, not three. Today you work out what is happening and get the room to one picture of it. Hale and the crew can hear every word you say.

BRIEFING 29W

A crewed lunar spacecraft reports an explosion, falling electrical power, and contradictory telemetry. The player takes the flight-director console and must first determine what can still be measured reliably.

WHAT YOU DECIDE 14W

Create a trusted state vector and a prioritized anomaly list before commanding the crew.

1.1 The spacecraft, or the sensors?

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · CHOICE

SCENE 53W

Nine minutes after the bang, three cabin pressure readings fall in the same second. Cabin pressure is how hard the air inside is pushing on the walls, and it drops when air leaks out. The dial gauge beside them, which is a needle on a spring rather than anything electrical, has not moved.

GUIDE 63W

Four options, and two facts have to be explained together. Three readings fell in the same second. The dial gauge did not move at all. Ask of each option how both of those could be true. The dial is a needle on a spring, so it shares nothing electrical with the other three. Three holes appearing at once is a lot to ask.

ASK 11W

What could make all three readings fall in the same second?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

several instruments can run off one shared circuit

VERDICT 89W

Three holes appearing in the same second is not a thing that happens. One fault behind all three is. So the question becomes what those three have in common, and the answer is the circuit they all run off. There is a check available, and it is the dial gauge. It measures the same cabin a completely different way, through a spring rather than through wires, and it has not moved. The leak listener hears nothing either. Together they say the cabin is fine and the wiring is not.

TAKEAWAY 16W

Three readings that fall together are one event. What they share is where to look first.

1.2 Build the first picture

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · SEQUENCE

SCENE 46W

Three consoles hold numbers that disagree. One of the two main power lines reads zero. The guidance computer, which keeps track of where the spacecraft is and which way it points, has restarted with no memory of what it saw. Nothing has been written down yet.

GUIDE 52W

All four steps will be done, so ask what each one needs to be true first. The recordings are wiped over within minutes. A neat answer worked out from two different clocks is still a wrong answer. And a command is only worth sending once you know how uncertain the picture is.

ASK 11W

Put the four steps in the order they have to happen.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a number without its units and its time is not a measurement

VERDICT 71W

The recordings from across the failure get wiped within minutes, so they are saved before anything is worked out. Units and clocks come next. A tidy answer built from mixed clocks is confidently wrong. Then the estimates, each with a range rather than one number. A number with no range cannot be argued with. And the only commands worth sending are the ones that help whatever turns out to be true.

TAKEAWAY 15W

An answer built to survive being wrong has to be built in a particular order.

1.3 The reading that cannot wait

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · CHOICE

SCENE 39W

Four screens are showing four problems at once. Camila Reyes wants to know which one the room should watch while the tracking pass — the few minutes when a dish on Earth can measure the spacecraft — comes round.

GUIDE 57W

All four readings matter and all four are being watched. Ask of each how fast it changes, and what happens to the crew if it goes wrong. Some of these get worse over hours. One of them gets worse in minutes, and there is nothing the crew can do without it. That is the difference being graded.

ASK 11W

Which reading tells you soonest whether the crew is in danger?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

people need air, warmth and power, and they need them on different clocks

VERDICT 95W

All four readings matter. What separates them is how fast each one runs out. Air is the quick one. A hole in the wall of a spacecraft empties the cabin in minutes, and there is no warning after that. The batteries have hours left in them. The cabin cools slowly, because heat leaks out through the walls rather than pouring out of a hole. So the flight director watches the pressure while the other three are still being counted. The order is set by the clock on each one, not by which number looks worst.

TAKEAWAY 18W

Air runs out in minutes. Warmth and power run out in hours, so the air is watched first.

Use units, vectors, graphs, measurement uncertainty, common-mode failure as an evidence chain rather than as isolated facts.

Find the Spacecraft

BEFORE THE SHIFT — TRIAL

Walk the floor before the first pass

The plot boards are at the front and the consoles step down towards them in four tiers. Walk the whole floor once, because when a number is wrong you have about ninety seconds to get to the desk that owns it.

PLAN CARD 73W

Nine hours after the bang. The one station still tracking sets in forty minutes. Camila Reyes, the guidance lead, will not sign a trajectory built on the computer that failed. Today you rebuild where the spacecraft is and how fast it is going. Dale Whitaker wants a number now, so the room can plan the correction. Reyes wants the right number. They are both right, and the clock is the same forty minutes.

BRIEFING 23W

Tracking data are intermittent, and the onboard computer has restarted. The team must reconstruct position and velocity from range, bearing, and timed observations.

WHAT YOU DECIDE 13W

Produce a consistent trajectory estimate and identify which new observation most reduces uncertainty.

2.1 Which measurement gives what

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · PROTOCOL

SCENE 32W

The onboard computer has restarted and the tracking comes and goes. The room has to rebuild where the spacecraft is from whatever can still be measured, and say what each measurement buys.

GUIDE 54W

Four things you need to know, and four ways to measure. Pair them by asking what each measurement actually notices. An angle? A time? A change of pitch? Or the difference between two earlier answers? One of the four is built out of the others, which is why it is the roughest of them.

ASK 13W

Match each thing you need to know to the measurement that gives it.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

saying where something is and how it is moving takes more than one number

VERDICT 75W

Each measurement answers a different question and none of them answers all of it. Photographing the spacecraft against known stars gives a direction and no distance. Timing a radio pulse out and back gives a distance and no direction. The pitch of the returning signal says how fast it is coming or going, and nothing about sideways motion. And rates come from repeating: two positions give a speed, two speeds give a change in speed.

TAKEAWAY 17W

No single measurement gives a position. Each one gives a piece, and the pieces are different shapes.

2.2 Rebuild where it is

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A PERSON · SEQUENCE

SCENE 30W

The computer restarted and lost its bearings. A distance, two angles and a pitch trace remain, recorded on different clocks. The next tracking pass will test whatever the room reconstructs.

GUIDE 51W

All four steps will happen. Ask of each what has to be sorted out first.

Measurements on different clocks cannot be fitted together until they are on one clock. A fit cannot be judged until you look at what it missed. And the burn happens at a set time, not now.

ASK 6W

Put the four steps in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

measurements only combine when they are on the same map and the same clock

VERDICT 72W

Everything goes onto one map and one clock first, or a tidy answer comes out wrong in a way nobody can see. The distance, the angles and the pitch then get fitted together. Each pins down something different, and fitted apart they give three answers. What the fit misses is read for pattern rather than size. A pattern is a force nobody allowed for. Then it is carried forward to the burn.

TAKEAWAY 13W

A useful answer predicts the next measurement, and says how sure it is.

2.3 What holds it in the curve

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A PERSON · CHOICE

SCENE 32W

The return path swings the spacecraft round the Moon at two and a half thousand kilometres up. Somebody on the loop has asked why it does not simply fall onto the surface.

GUIDE 57W

All four options offer something holding the spacecraft up. Ask of each whether it names a real push. Two of them invent one. There is nothing out there for a wing to push against. Gravity is the only force here, and it points inward. So ask what the sideways speed does to something that is already falling.

ASK 14W

Why does the spacecraft need any speed at all to stay in that path?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

something moving in a circle is being pulled toward the middle, even at a steady speed

VERDICT 74W

It is falling. It has been falling the whole way round. What stops it arriving is sideways speed: while gravity pulls it down, it moves along, and the Moon's surface curves away underneath at about the same rate. Too slow and the path meets the ground. Too fast and it swings out and away. Nothing is holding it up, which is exactly why the crew float. Falling and floating are the same thing here.

TAKEAWAY 14W

An orbit is falling, with enough sideways speed that the ground keeps curving away.

END OF THE SHIFT 18W

Use position, velocity, acceleration, coordinate systems, graphs of motion as an evidence chain rather than as isolated facts.

Turn Without Wasting Fuel

BEFORE THE SHIFT — FOLLOW

Duarte walks a number from the aerial to the board

A reading comes down from the spacecraft, through three desks, and onto the wall. Duarte walks the whole path once. Stay with her, and watch which desk is allowed to change it.

PLAN CARD 71W

Twenty-six hours in. The guidance computer is off to save power and one set of thrusters is dead. Hale has to turn the spacecraft ninety degrees by hand. Today you agree what the turning numbers mean and write the steps for him. Every degree costs fuel twice, once to start it and once to stop it. Let it tumble and they lose the antenna, the power and the engine burn together.

BRIEFING 20W

The main guidance platform is unavailable, and the crew must reorient the spacecraft using limited thrusters and visible reference objects.

WHAT YOU DECIDE 13W

Develop a controlled reorientation procedure that stops the rotation at the required attitude.

3.1 How a spacecraft turns

ROTATION & STRUCTURES · STRUCTURES TEST BAY · A A PLACE · PROTOCOL

SCENE 27W

The guidance computer is out and the crew is about to fly by hand. Before anybody touches a thruster, the room says plainly what each change does.

GUIDE 61W

Four changes on the left, four results on the right. Pair them by asking what each change does to two things. How hard is it being turned? And how hard is it to turn? One of the four has nothing pushing from outside at all, which is what makes it the odd one. The crew will feel every one of these.

ASK 10W

Match each change to what it does to the turning.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

a push off to one side is what makes something start or stop turning

VERDICT 80W

Four rules cover every hand-flown turn. The same push further from the middle turns harder, which is the one the crew can actually choose. The same push on a heavier, more spread-out craft turns it more slowly, and stops it more slowly too. A push one way and an equal push back cancels out, which is what starting and stopping a turn actually is. And moving things about inside changes nothing overall, because nothing inside can push on the outside.

TAKEAWAY 14W

Turning follows the same rules as moving. It is just harder to see them.

3.2 Turning it by hand

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · FLY

SCENE 33W

The guidance computer is off. The crew has to turn the spacecraft ninety degrees using two thruster pairs and a mark on the window. Every extra correction spends fuel meant for the return.

GUIDE 57W

You set two things: how hard to start the turn, and where to start braking. Then the plan runs and you watch it. Nothing slows this turn down by itself, so whatever you start you have to stop on purpose. The braking has to come early rather than at the mark, and every extra correction spends fuel.

ASK 17W

How hard do you start the turn, and where do you start braking so it arrives still?

BEHIND THE BUTTON 154W

Why the brake has to come early. There is no air and no friction in space, so a spacecraft that starts turning keeps turning at the same rate. It stops only when you push it the other way by the same amount. Firing at the mark means arriving still turning, and then having to fix it.

What the window mark tells you. It shows which way the spacecraft is pointing. It does not show how fast it is turning, so you have to judge that from how quickly the view slides past. That is why a gentle start is easier to fly than a hard one, even though it takes longer.

Why the fuel matters more than the minutes. The fuel being spent is what the return needs later. A fast turn with three corrections can cost more than the time it saved is worth, and there is no way to get it back.

TAKES AS READ

a push changes how fast something turns · it keeps turning after the thruster stops

VERDICT 84W

A thruster pulse does not make a turn that stops itself. It changes how fast the spacecraft is turning, and when the pulse ends it keeps going at that rate. Stopping needs the opposite thruster, and that takes just as long as starting did. So the braking has to begin before the target, by about the angle the starting pulse used up. Brake at the target and it sails past. The turn only works if the angle and the turning both reach zero together.

TAKEAWAY 15W

Stopping the push does not stop the motion, so the braking has to start early.

3.3 Turning effect from a thruster

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · BALLPARK

SCENE 31W

A thruster pushes with two hundred newtons, three metres out from the middle of the spacecraft. The crew needs the turning effect before it can predict how fast the turn starts.

GUIDE 50W

Five numbers, and three belong elsewhere. The weight of the spacecraft, the docking ring, and a half. Ask of each whether the turning depends on it. Then note what the crew can change. Picking a different thruster changes how far out the push acts, without changing the thruster at all.

ASK 8W

How much turning effect does the thruster give?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

the same push further from the middle turns something harder

VERDICT 77W

Turning effect is the push multiplied by how far out it acts. Two hundred newtons, three metres from the middle, is six hundred. Move the same thruster in to half the distance and the turning effect halves, with the same fuel burnt. That is why the crew care which pair fires. They cannot make the thruster stronger, and they can choose one that acts further out, and it is exactly the same amount of fuel either way.

TAKEAWAY 13W

Where a push is applied matters as much as how hard it pushes.

3.4 The spacecraft, or the sensor

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · BELT

SCENE 37W

The console has flagged more numbers than the team can look at before handover. Some of them are the spacecraft doing something. The rest are the sensor going wrong, and both are just numbers on a list.

GUIDE 46W

Two bins. Ask whether anything else on board agrees. A real change shows up in more than one place and happens at a speed that is possible. A broken sensor jumps instantly, goes off the end of its scale, or sits still while everything else moves.

ASK 12W

Send each flag to the bin that says what the problem is.

BEHIND THE BUTTON 75W

Why this comes first. The answer to the spacecraft misbehaving is to do something. The answer to a sensor misbehaving is to stop trusting the number. Getting it the wrong way round spends fuel on a problem that is not there.

What a broken sensor looks like. A jump nothing could really do. A reading past the end of the scale. A number that has not moved in an hour while everything near it has.

TAKES AS READ

a number comes from a sensor, and a sensor can break

VERDICT 136W

Every flag is a number that moved. A number is not the thing itself. When the spacecraft really is warming up, it shows in more than one place. The structure warms. The radiator warms. The power draw agrees. And it all happens at a speed heating can manage. A broken sensor does none of that. It jumps all at once. Or it parks past the end of its range. Or it freezes while everything near it changes. Or it disagrees with two other sensors measuring the same thing. So the test is whether anything else agrees, not whether the number looks sensible. A broken sensor can give a very sensible number. The cost runs both ways. Chasing a broken sensor wastes fuel on a problem nobody has. Calling a real problem a bad reading is worse.

TAKEAWAY 10W

The spacecraft and the sensor go wrong in different ways.

END OF THE SHIFT 21W

Use torque, angular acceleration, moment of inertia, angular momentum, center of mass as an evidence chain rather than as isolated facts.

The Power Budget

PLAN CARD 68W

Thirty-four hours in. Priya Shah has the power reserve down to hours, and she wants everything not vital switched off today. Aaron Brooks says the two things on his list are the air scrubber and the heaters. His crew are sleeping in everything they own. Today you work out how long the batteries last and choose what goes off. Whatever you switch off, somebody up there feels it.

BRIEFING 19W

Fuel cells and batteries are degraded. Every electrical load now competes with guidance, communications, thermal control, and life support.

WHAT YOU DECIDE 13W

Build a load-shedding plan that preserves essential functions through the next critical maneuver.

4.1 How long can the battery last?

POWER & CIRCUITS · POWER & CIRCUITS BAY · A A PLACE · BALLPARK

SCENE 43W

The fuel cells are gone, and the crew is on the entry batteries days earlier than any plan allowed. The flight doctor, who watches the crew's health from the ground, wants a number of hours before the next decision about switching things off.

GUIDE 57W

Five numbers, and three belong to other questions. The voltage, the brief peak, and the time to the next burn. Ask of each whether it is an amount stored, or a rate of use. An amount divided by a rate leaves a time. And note what the answer assumes: nothing switching on, and every last bit usable.

ASK 6W

How long will the batteries last?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a store is an amount, and a draw is how fast it is being used

VERDICT 82W

There is a store and there is a draw, and dividing one by the other gives hours. Eighteen kilowatt hours at three kilowatts is six hours. It is an ideal figure. It assumes the draw stays flat and that every last bit of the store is usable, and neither is quite true. A cold battery gives less than its label. It is still the number every argument this shift is measured against, so it is worth having early and worth calling rough.

TAKEAWAY 20W

An amount divided by a rate is a time, and that time is what every other decision gets argued against.

4.2 Why is the voltage sagging?

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · CHOICE

SCENE 32W

A main power line is sagging at twenty-four volts instead of twenty-eight, while the current drawn from it sits where it was planned to be. Engineers want to start pulling equipment off.

GUIDE 60W

Four options, and three facts have to fit together. The supply is fine. The current is where it was planned. Something is hot. Ask of each option how many of those it covers. A hidden extra load would show in the current. A wrong meter would not warm anything up. Equipment is about to be pulled off on this answer.

ASK 10W

Which fault fits the voltage, the current and the heat?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

current pushing through something that resists it drops the voltage and makes heat

VERDICT 62W

Three readings, one fault, and each says something different. Something extra drawing power would show up as more current than planned. The current is where it should be. A failed supply would show as less coming in, and what comes in is normal. What is left is something in the path resisting. It eats voltage going past and turns it into heat.

TAKEAWAY 16W

Voltage, current and heat are three views of one fault, and one view is not enough.

4.3 Switch things off without losing the mission

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · ALLOCATE

SCENE 41W

Forty-one amp-hours left, and sixty-two hours to fly. An amp-hour is one hour of giving one amp, so forty-one of them is not much. Guidance, radio, heating and life support all want power. Something has to be turned down within minutes.

GUIDE 54W

Forty-one amp-hours, sixty-two hours to fly, and everything switched on uses more than that. Watch the list of questions rather than the loads: it shows what the spacecraft can still do with your plan. Every hour you give one thing is an hour taken from another, and something has to come down within minutes.

ASK 9W

How many hours of each can the spacecraft afford?

BEHIND THE BUTTON 130W

How little there is. At the current draw the whole reserve lasts under ten hours, and there are sixty-two to fly. So this is not a matter of trimming a bit. The draw has to come down by several times over and stay down.

Why the loads are not equal. The radio and the guidance computer can be switched on for a while and off again. Heating changes slowly, so it can be cycled. Life support cannot be interrupted at all. That is what makes a plan possible.

Why the first cut cannot wait. Power is being used at the current rate while you decide, so a slow decision has already spent some of what it was deciding about. Cutting something early buys the time to plan the rest properly.

TAKES AS READ

an amp-hour is a current drawn for an hour · some things can be switched on and off, and some cannot

VERDICT 68W

This is not a list of what matters most. It is a sum. Every load costs its draw multiplied by the hours it runs, and the total cannot pass forty-one. Life support runs the whole way, so it is bought first. Guidance is needed in fixed windows, so those get booked. Heating can be switched on and off, and not removed, because a cold battery gives less back.

TAKEAWAY 21W

A power budget is not a ranking of what matters. It is hours multiplied by draw, and the total is fixed.

4.4 Which measurement gives what

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · CALLBACK · PROTOCOL

SCENE 32W

The onboard computer has restarted and the tracking comes and goes. The room has to rebuild where the spacecraft is from whatever can still be measured, and say what each measurement buys.

GUIDE 54W

Four things you need to know, and four ways to measure. Pair them by asking what each measurement actually notices. An angle? A time? A change of pitch? Or the difference between two earlier answers? One of the four is built out of the others, which is why it is the roughest of them.

ASK 13W

Match each thing you need to know to the measurement that gives it.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

saying where something is and how it is moving takes more than one number

VERDICT 75W

Each measurement answers a different question and none of them answers all of it. Photographing the spacecraft against known stars gives a direction and no distance. Timing a radio pulse out and back gives a distance and no direction. The pitch of the returning signal says how fast it is coming or going, and nothing about sideways motion. And rates come from repeating: two positions give a speed, two speeds give a change in speed.

TAKEAWAY 17W

No single measurement gives a position. Each one gives a piece, and the pieces are different shapes.

Use current, voltage, resistance, electrical power, energy budget as an evidence chain rather than as isolated facts.

The Cabin Is Cooling

BEFORE THE SHIFT — HUNT

Six spare tapes, and the log lists four

Every pass is recorded twice, and two of the spare tapes are not where the log says. A tape nobody can find is a pass nobody can play back, and you only get the pass once.

PLAN CARD 78W

Two days after the bang. Brooks won the argument, and it cost Shah her margin. The cabin is at four degrees and falling. Water is running down the walls, and Ruiz can no longer write clearly. Today you work out how fast the heat is escaping and choose what to wrap. There is power to fight it one way, not three. Brooks is arguing for the crew and Shah for the batteries, and they want the same watts.

BRIEFING 23W

With most systems powered down, cabin temperature falls and condensation forms. The player must balance conduction, convection, radiation, metabolic heat, and battery performance.

WHAT YOU DECIDE 14W

Create a thermal survival plan that protects crew, electronics, and batteries with minimal power.

5.1 How much does the cabin cool?

THERMAL & LIFE SUPPORT · LIFE SUPPORT LAB · A A PLACE · BALLPARK

SCENE 35W

With most things switched off, the cabin is losing about a kilowatt more heat than it makes. The flight doctor wants to know how long the crew has before the cold becomes a medical problem.

GUIDE 55W

Four numbers, and two of them belong elsewhere. The temperature now, and the hours, are not in this sum. Ask of each whether it is energy lost, or energy for one degree. And note what the answer means. The cabin cools slowly, which buys hours now and costs the same energy to warm back up.

ASK 9W

How far does the cabin cool over three hours?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

it takes energy to warm something up, and the same energy leaves as it cools

VERDICT 88W

A kilowatt is a thousand joules leaking away every second. Three hours is about eleven thousand seconds, so the cabin loses roughly 11 million joules. That sounds enormous. It is not, because the cabin and everything in it is heavy, and heavy things take about 12 million joules to shift by a single degree. That is the useful part. The cold is real and it is slow, so the crew has hours rather than minutes. The same fact is why warming it back up later costs so much.

TAKEAWAY 12W

Big things change temperature slowly, and that slowness is time to think.

5.2 How the heat is getting out

THERMAL & LIFE SUPPORT · LIFE SUPPORT LAB · A PERSON · PROTOCOL

SCENE 32W

The cabin is cooling and the crew wants to do something about it. What they can do depends on how the heat is actually leaving, so the room names the route first.

GUIDE 54W

Four things happening in the cabin, and four ways heat moves. Pair them by asking what is carrying the warmth. Touching? Moving air? Or nothing at all? One of these needs nothing in between. The pairing decides what a fix looks like, because wrapping, fans and shiny surfaces each work on a different route.

ASK 13W

Match each thing happening in the cabin to how the heat is moving.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

heat moves by touching, by riding on moving air, or by beaming across a gap

VERDICT 81W

Heat leaves three ways, and each has a different fix. By touching, which needs contact and is short range. By riding on moving air, which is why circulation spreads warmth about. And by beaming straight across a gap, which needs nothing in between and works fine in vacuum. A metal strut running out to the cold structure is the worst kind, because it touches all the way and never stops. Insulation, circulation and shiny surfaces each work on a different one.

TAKEAWAY 13W

Naming the route is what tells you which fix would work at all.

5.3 Keep the cabin survivable

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · CHOICE

SCENE 33W

The cabin is at four degrees and falling. Water is beading on the walls and the crew is sleeping in their suits. Heating all of it is not affordable on what is left.

GUIDE 52W

All four options spend the same fixed energy. Ask of each whether it slows the loss, or puts back heat that has already gone. Wrapping does the first and cannot do the second. Then ask where cold costs most. Cold batteries give less, so a cold problem turns into a power problem.

ASK 16W

Four degrees and falling, on a fixed amount of energy. What do you spend it on?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

slowing a loss and replacing it are different things

VERDICT 77W

There are two ways to fight cold and they cost differently. Replacing heat means paying again the moment you stop. Slowing the loss keeps working after the spending stops, which is what matters when the energy is fixed. So it goes round the crew and round the batteries, where cold does the most damage. Cold batteries also give back less than warm ones, so a heat problem quietly becomes a power problem if nobody attends to it.

TAKEAWAY 14W

Slowing a loss keeps paying. Replacing it has to be paid again every time.

5.4 Cabin pressure, through the work period

THERMAL & LIFE SUPPORT · LIFE SUPPORT LAB · A PERSON · HOLD

SCENE 37W

The crew are working in the middle module and the cabin has to hold its pressure while they do. An airlock cycles, the air cleaner comes on, and the small leak on the log is still there.

GUIDE 46W

Hold the cabin pressure inside the band on the console. The band narrows later, because the crew move towards the module with the suspect seal. A leak keeps leaking, so set the makeup valve to match it rather than opening it after the gauge has fallen.

ASK 10W

Hold the cabin inside the band through the work period.

BEHIND THE BUTTON 64W

Why a leak is not one drop. Gas escapes for as long as the hole is open, so the pressure keeps falling. Opening a valve once buys a few minutes and changes nothing.

Why the airlock is different. Cycling it takes a fixed amount of gas out, once. That wants a correction rather than a new setting, and both show on the same gauge.

TAKES AS READ

a sealed space loses pressure if gas escapes

VERDICT 143W

A leak is a rate, and everything else follows from that. Gas escapes for as long as the hole is open, so the pressure keeps falling and never settles. A valve opened once and shut again buys a few minutes. What works is setting the makeup flow to match the loss and leaving it. That also measures the leak: the flow you need to hold steady is the leak rate, and that is the number the ground needs to decide whether the module can keep being used. The airlock is the opposite. Cycling it takes a fixed amount of gas out, once, so it wants a correction rather than a new setting. Telling those two apart on one gauge is the skill. The band narrows because the crew are moving towards the module with the suspect seal, where the same drop matters more.

TAKEAWAY 16W

A leak is a rate, so it is answered by a setting rather than a correction.

5.5 Rebuild where it is

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · CALLBACK · SEQUENCE

SCENE 30W

The computer restarted and lost its bearings. A distance, two angles and a pitch trace remain, recorded on different clocks. The next tracking pass will test whatever the room reconstructs.

GUIDE 51W

All four steps will happen. Ask of each what has to be sorted out first.

Measurements on different clocks cannot be fitted together until they are on one clock. A fit cannot be judged until you look at what it missed. And the burn happens at a set time, not now.

ASK 6W

Put the four steps in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

measurements only combine when they are on the same map and the same clock

VERDICT 72W

Everything goes onto one map and one clock first, or a tidy answer comes out wrong in a way nobody can see. The distance, the angles and the pitch then get fitted together. Each pins down something different, and fitted apart they give three answers. What the fit misses is read for pattern rather than size. A pattern is a force nobody allowed for. Then it is carried forward to the burn.

TAKEAWAY 13W

A useful answer predicts the next measurement, and says how sure it is.

END OF THE SHIFT 17W

Use heat capacity, conduction, convection, radiation, energy balance as an evidence chain rather than as isolated facts.

The Air Problem

PLAN CARD 78W

Fifty-eight hours in. Carbon dioxide is climbing in a cabin built for two and holding three. Petra Novak has been on the radio with the crew for an hour. The spare canisters are the wrong shape for the socket. Today you find out why the air is not being cleaned, and build something out of what is aboard. Hale has started making small mistakes reading numbers back. That is the first sign, and he cannot see it himself.

BRIEFING 24W

Carbon dioxide removal is failing, and pressure readings differ between compartments. The player must connect gas laws, partial pressure, fluid flow, and filter resistance.

WHAT YOU DECIDE 16W

Develop a safe temporary air-cleaning and circulation strategy using only validated physical principles and monitored limits.

6.1 Follow the air

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · CHAIN

SCENE 46W

Carbon dioxide keeps climbing. The scrubber, which is the machine that takes that gas back out of the air, has a fan that sounds normal and draws its usual current. The airflow through it is low. There are pressure readings on both sides of the filter.

GUIDE 58W

Build the path the air takes, in order, then name the link where it becomes the problem. The fan sounds normal and draws its usual current, which tells you the fan is fine. The pressure readings on both sides of the filter are what show you where the blockage is, so read those instead of trusting the sound.

ASK 8W

Where does the air path become the problem?

BEHIND THE BUTTON 135W

Why a working fan and low airflow means a blockage. A fan uses more current the more air it moves. Meeting a blockage, it moves less air, works less hard, and sounds completely normal. So the current reading is evidence about the fan, not about the system.

What a pressure difference tells you. Pushing air through a blockage costs pressure. A big drop across the filter and a small one across the scrubber says the filter is blocked. The other way round says the scrubber is. Those need different repairs.

Why the carbon dioxide is the clock. It climbs whenever the scrubber removes it more slowly than the crew makes it, and people get slow and confused well before the level is dangerous. The problem has to be found while everybody can still think clearly.

TAKES AS READ

a motor running does not prove any air is moving · air has to pass through every part of a loop

VERDICT 67W

The fan makes a push. How much air moves depends on everything downstream of it. Normal current only proves the fan has power. The pressure difference across the filter is large and the flow is small, which is what a blockage looks like. Every breath has to get through that filter before it reaches anything that cleans it. So the loop runs at whatever the filter allows.

TAKEAWAY 18W

A chain moves at the rate of its most restricted link, not the rate of its noisiest one.

6.2 How much they breathe out

THERMAL & LIFE SUPPORT · LIFE SUPPORT LAB · A A PLACE · BALLPARK

SCENE 32W

Three crew, each breathing out about twenty litres of carbon dioxide an hour. The workshop needs to know what a home-made scrubber has to keep up with over the next six hours.

GUIDE 55W

Five numbers, and two belong to a different question. A whole day, and a figure by weight rather than by volume. Ask of each whether it is a rate, a count, or a length of time. What comes out is the least a scrubber has to keep up with. Anything slower only slows the rise.

ASK 10W

How much carbon dioxide do they make in six hours?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a rate multiplied by a time gives a total

VERDICT 75W

Three people at twenty litres an hour make sixty litres an hour between them. Six hours of that is three hundred and sixty. That number is the bar. Anything that removes carbon dioxide more slowly than the crew makes it does not hold the level steady. It only makes the rise slower. The workshop does not need this precise. It needs to know whether what they can build is big enough to matter at all.

TAKEAWAY 13W

The rate something arrives at sets the smallest rate that can remove it.

6.3 Make the air breathable

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · CHOICE

SCENE 32W

Carbon dioxide is close to the limit and still climbing. The spare canisters are square, the socket is round, and there is tape, a suit hose, a sock and a plastic cover.

GUIDE 55W

All four options describe something the fix could be judged on. Ask of each whether it changes how much air really goes through the canister. Air that slips round the edge comes out unchanged. A big canister with a gap does worse than a small one taped tight. The fan sounds the same either way.

ASK 17W

Tape, a hose, a sock and a plastic cover. What does the fix have to get right?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

air, like water, goes the easiest way it is offered

VERDICT 81W

Two things have to be true, and only one of them is obvious. There has to be something that takes the carbon dioxide out, and every bit of air has to be made to go through it. Air is lazy. Leave a gap round the canister and most of it goes round instead of through, and comes back into the cabin exactly as it left. A big canister with a gap round it cleans less than a small one taped tight.

TAKEAWAY 12W

Nothing gets cleaned unless it goes through the thing that cleans it.

6.4 What holds it in the curve

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · CALLBACK · CHOICE

SCENE 32W

The return path swings the spacecraft round the Moon at two and a half thousand kilometres up. Somebody on the loop has asked why it does not simply fall onto the surface.

GUIDE 57W

All four options offer something holding the spacecraft up. Ask of each whether it names a real push. Two of them invent one. There is nothing out there for a wing to push against. Gravity is the only force here, and it points inward. So ask what the sideways speed does to something that is already falling.

ASK 14W

Why does the spacecraft need any speed at all to stay in that path?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

something moving in a circle is being pulled toward the middle, even at a steady speed

VERDICT 74W

It is falling. It has been falling the whole way round. What stops it arriving is sideways speed: while gravity pulls it down, it moves along, and the Moon's surface curves away underneath at about the same rate. Too slow and the path meets the ground. Too fast and it swings out and away. Nothing is holding it up, which is exactly why the crew float. Falling and floating are the same thing here.

TAKEAWAY 14W

An orbit is falling, with enough sideways speed that the ground keeps curving away.

END OF THE SHIFT 19W

Use ideal gases, partial pressure, flow resistance, pressure difference, filtration as an evidence chain rather than as isolated facts.

Communication Fades

BEFORE THE SHIFT — CANVASS

Which desks saw the drop, and when

Something in the power reading dipped for four seconds. Whether every desk saw it at the same moment decides if it was the spacecraft or the ground station. Ask round the room until enough desks have answered.

PLAN CARD 69W

Sixty-six hours in. The radio went weak the moment the spacecraft turned. Hale is answering calls he can barely hear. Naomi Ito wants the aim checked first. Her network officer wants the receiver checked instead. Today you find the cause and get the link back. Losing the radio costs twice, because it is also how the room measures where the spacecraft is. There is still a burn to fly.

BRIEFING 25W

The spacecraft signal weakens and shifts in frequency as geometry changes. The team must separate transmitter failure from antenna pointing, range loss, and Doppler effects.

WHAT YOU DECIDE 14W

Restore a reliable low-bandwidth link and use the signal itself as a navigation measurement.

7.1 Why did the signal fade?

WAVES & COMMUNICATIONS · CAPSULE COMMUNICATIONS · A A PLACE · CONTROL

SCENE 48W

Voice and data have gone weak at two ground stations at once. A ground station is a dish antenna on Earth, and these two are on opposite sides of the world. The spacecraft's transmitter is putting out what it should. Three things can be swapped without risking anything.

GUIDE 78W

The number you watch is how strong the signal is. Right now it is about twelve decibels low at both dishes. Each test changes one part of the path and can be undone. A change counts only if the number moves more than it wanders on its own. Both dishes are weak at once, which already tells you the cause has to be something they share. Change one thing, undo it, and name the part the fade follows.

ASK 8W

Which change makes the fade appear and disappear?

BEHIND THE BUTTON 153W

Why two dishes at once is a clue. The two ground stations sit on opposite sides of the Earth. They have their own receivers, their own crews and their own weather. A fault in one cannot make the other weak. So the cause has to be something both of them share, which means the spacecraft end or the aim.

What decibels are. A decibel is a way of writing a ratio. Three decibels is about half the power. Ten decibels is a tenth of it. Twelve decibels is about one sixteenth. That is the difference between a voice you can work with and one you cannot.

Why every test can be undone. There are people aboard the spacecraft. A change you cannot undo is a decision, not a test. All three of these can be tried and put back, which is what lets you check that the fade follows the change both ways.

TAKES AS READ

a fair test changes one thing and leaves the rest alone

VERDICT 79W

Several things can weaken a radio link, and only one test settles which. Swap to a different ground receiver and nothing changes, so it is not one receiver. Swap the transmitter chain and nothing changes, so it is not the transmitter. Correct where the antenna is pointing and almost all the missing signal comes back. Point it wrong again and the fade returns. Following the change in both directions is what makes it a cause rather than a coincidence.

TAKEAWAY 12W

A cause is proved when the effect follows the change both ways.

7.2 How long is one wave

WAVES & COMMUNICATIONS · CAPSULE COMMUNICATIONS · A PERSON · BALLPARK

SCENE 26W

The link is fading, and the antenna team needs the length of one wave for its pointing model. The spacecraft transmits two billion waves a second.

GUIDE 58W

Four numbers, and two of them are traps. One is the same figure misread as millions rather than billions. One is the speed of sound, which has nothing to do with radio. Ask of each whether it belongs to this wave. And note what the answer is for: it sets how carefully the dish has to be aimed.

ASK 5W

How long is one wave?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

radio travels at a fixed speed, three hundred million metres every second

VERDICT 64W

Radio covers three hundred million metres in a second. In that same second, two billion waves go past. So one wave is that distance shared between them, which is about fifteen centimetres. The number matters because the dish sends its signal in a beam. Short waves make a narrow beam. A narrow beam reaches further, and it has to be aimed far more carefully.

TAKEAWAY 13W

How long a wave is decides how fussy the aiming has to be.

7.3 Get the link back

ROTATION & STRUCTURES · STRUCTURES TEST BAY · A A PLACE · SEQUENCE

SCENE 25W

The signal is weak and drifting. The crew is answering calls it can barely hear. The next navigation update depends on getting the link back.

GUIDE 59W

All four moves will be made, so ask which buys most for least. Where the spacecraft points is the biggest thing in the whole link, and it costs nothing to check. Slowing the message down buys clarity without touching the spacecraft. And one of these four is not a repair at all. It is what the repair is for.

ASK 6W

Put the four moves in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a slower message needs less signal to be read

VERDICT 65W

Pointing is the biggest thing on the list and the cheapest to fix, so it goes first.

Slowing the message down comes next. Less is being sent, so less signal is needed to read it, and nothing aboard has to change. Then the big ground dishes get told where to listen, because the signal has drifted. What comes back feeds the navigation the burn needs.

TAKEAWAY 14W

Do the biggest and cheapest thing first, and let each move buy the next.

7.4 The spacecraft, or the sensors?

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · CALLBACK · CHOICE

SCENE 53W

Nine minutes after the bang, three cabin pressure readings fall in the same second. Cabin pressure is how hard the air inside is pushing on the walls, and it drops when air leaks out. The dial gauge beside them, which is a needle on a spring rather than anything electrical, has not moved.

GUIDE 63W

Four options, and two facts have to be explained together. Three readings fell in the same second. The dial gauge did not move at all. Ask of each option how both of those could be true. The dial is a needle on a spring, so it shares nothing electrical with the other three. Three holes appearing at once is a lot to ask.

ASK 11W

What could make all three readings fall in the same second?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

several instruments can run off one shared circuit

VERDICT 89W

Three holes appearing in the same second is not a thing that happens. One fault behind all three is. So the question becomes what those three have in common, and the answer is the circuit they all run off. There is a check available, and it is the dial gauge. It measures the same cabin a completely different way, through a spring rather than through wires, and it has not moved. The leak listener hears nothing either. Together they say the cabin is fine and the wiring is not.

TAKEAWAY 16W

Three readings that fall together are one event. What they share is where to look first.

Use waves, frequency, wavelength, inverse-square behavior, Doppler effect as an evidence chain rather than as isolated facts.

The Structure Is Vibrating

BEFORE THE SHIFT — EVADE

Keep clear of the press desk before the pass

Somebody has been waiting by the door for a number since six, and nothing goes out before it is checked twice. Get to your console without going past them, and leave the talking until after the pass.

PLAN CARD 71W

Ninety hours in, and Chidi Okoye caught this one early. Nothing else this week has been caught early. A panel shakes whenever a pump runs near one speed. A pipe behind it has been rubbing for an hour. Today you work out what is shaking it, and stop it. The shaking dies away as soon as the pump moves off that speed. This is a day where something is simply fixed.

BRIEFING 26W

A pump and flexible panel produce a growing oscillation near one operating speed. The player must identify resonance and protect the structure without disabling life support.

WHAT YOU DECIDE 15W

Move the system away from resonance and verify that the mitigation works across operating conditions.

8.1 What is shaking it?

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · CHOICE

SCENE 32W

A bad vibration only appears when the wheel runs near one speed. Above and below that it dies away. Two sensors on different parts of the structure see the same narrow band.

GUIDE 55W

Four options, and three facts have to fit. It only happens near one speed. It dies away either side. Two sensors on different parts see the same thing. Ask of each option how many of those it explains. A loose sensor would not be backed up by a second one. Something hitting it happens once.

ASK 5W

Which explanation fits the readings?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

everything has speeds at which it prefers to shake

VERDICT 80W

Everything has a rhythm it likes to shake at, the way a swing does. Push a swing at any old moment and little happens. Push it in time with itself and it goes higher every push. Here the wheel is the push, and one speed lands in time with the structure's own rhythm. Move off that speed either way and the pushes fall out of step. That is why the vibration has a narrow band rather than growing with speed.

TAKEAWAY 16W

What matters is not how hard it shakes, but what is shaking it, and how fast.

8.2 How far to stay clear

ROTATION & STRUCTURES · STRUCTURES TEST BAY · A A PLACE · BALLPARK

SCENE 34W

The bad band runs from three thousand one hundred to three thousand three hundred turns a minute. The rule is to stay two hundred clear of it, and the pump still has to run.

GUIDE 56W

Four numbers, and two of them are the top of the band and the worst point in it. Those belong to the other side of the problem. Ask which two the question needs. Below the band means below its bottom edge, with the clearance taken off. Going the wrong way puts the pump inside the band.

ASK 11W

What is the highest speed that is safely below the band?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a shaking problem has a band of speeds, with room either side

VERDICT 80W

A resonance is a band rather than a point, and a margin either side of it is what stops the pump wandering back in. The band starts at three thousand one hundred. Two hundred clear of that is two thousand nine hundred, and that is the fastest the pump can safely run below it. Running slower is fine. Running anywhere between three thousand one hundred and three thousand three hundred is not, and neither is sitting just outside and drifting.

TAKEAWAY 14W

Avoiding a resonance is arithmetic on the band, not a matter of running gently.

8.3 Stop the shaking

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · CHOICE

SCENE 32W

A panel shakes whenever the pump runs near one speed, and it has grown worse over three cycles. The pump is needed, and its speed can still be changed from the panel.

GUIDE 63W

Shaking like this needs two things. A structure with its own rhythm, and something pushing in time with it. All four options act on one of those, or on neither. Ask which, and how fast the crew could do it. It has already grown worse over three cycles. So one option's bet, that it holds together on the way through, is not safe.

ASK 11W

Which change stops the pushes landing in time with the panel?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

shaking like this needs a structure and something pushing it in time

VERDICT 80W

This kind of shaking needs two things: a structure with a rhythm, and something pushing in time with it. In flight, changing what is pushing is far easier than changing the structure. Moving the pump speed off the band stops the pushes landing in time, and it can be undone in a second. Deadening the panel is the proper permanent fix, and it needs hardware and somebody's hands on it. Running faster through the band assumes it survives the crossing.

TAKEAWAY 12W

Change what is pushing, when you cannot change what is being pushed.

8.4 What the room is answering now

WAVES & COMMUNICATIONS · CAPSULE COMMUNICATIONS · A A PLACE · SPOT

SCENE 24W

Calls come in faster than they can all be answered. One rule says which get answered first, and it changes as the situation does.

GUIDE 44W

Answer what the rule asks for and hold the rest. The rule is said once and then it stands. What counts is the calls either side of a change, because they show whether the room is working to this rule or the last one.

ASK 12W

Answer to the rule in force, and keep listening for the rule.

BEHIND THE BUTTON 67W

Why the rule changes. Before a burn everything waits for the burn. Once a leak is confirmed everything waits for the leak. When the crew are asleep it changes again, because waking them costs something.

Why the change is the dangerous part. Two more minutes on the old rule gives the loop to a call that no longer matters, while the one that does waits behind it.

TAKES AS READ

a control room answers calls in an order somebody sets

VERDICT 102W

Three rules run across the shift and each wants different calls. Anything about the burn, then anything about the leak, then anything that would wake the crew. A call can match two at once, and that is what makes a change cost something. What the panel scores is the calls either side of a change. Most of what comes in is wanted by neither rule, so holding those proves nothing. The room's own version is the two minutes after a leak is confirmed. The rule has changed, and the loop is still going to the people who had it a moment ago.

TAKEAWAY 18W

The cost of a rule that has changed is paid by whoever is still using the old one.

8.5 Turning effect from a thruster

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · CALLBACK · BALLPARK

SCENE 31W

A thruster pushes with two hundred newtons, three metres out from the middle of the spacecraft. The crew needs the turning effect before it can predict how fast the turn starts.

GUIDE 50W

Five numbers, and three belong elsewhere. The weight of the spacecraft, the docking ring, and a half. Ask of each whether the turning depends on it. Then note what the crew can change. Picking a different thruster changes how far out the push acts, without changing the thruster at all.

ASK 8W

How much turning effect does the thruster give?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

the same push further from the middle turns something harder

VERDICT 77W

Turning effect is the push multiplied by how far out it acts. Two hundred newtons, three metres from the middle, is six hundred. Move the same thruster in to half the distance and the turning effect halves, with the same fuel burnt. That is why the crew care which pair fires. They cannot make the thruster stronger, and they can choose one that acts further out, and it is exactly the same amount of fuel either way.

TAKEAWAY 13W

Where a push is applied matters as much as how hard it pushes.

END OF THE SHIFT 19W

Use simple harmonic motion, natural frequency, resonance, damping, forced oscillation as an evidence chain rather than as isolated facts.

The Last Correction

PLAN CARD 75W

A hundred and six hours in, and it is Carter's turn. She has waited four days for it. Several ground stations suddenly put the spacecraft ahead of its path. They agree with each other, and they all run off the same clock. Whitaker wants the burn now. Today you decide whether the spacecraft really moved. Carter's rule is that a correction is only worth making when it is bigger than the doubt in the measurement.

BRIEFING 25W

Hours before entry, tracking stations disagree slightly. The player must combine measurements with correlated errors and decide whether another burn reduces or increases total risk.

WHAT YOU DECIDE 14W

Decide whether to burn using the expected benefit relative to navigation and propulsion uncertainty.

9.1 Really off course, or one bad clock?

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · TRACE

SCENE 28W

Several ground measurements suddenly put the spacecraft ahead of where it should be. They agree with each other. A star camera and the clock records are both available.

GUIDE 49W

Open each measurement to see what it depended on. Keep the ones that would still stand if one clock were wrong, and untick the rest. Then name what the others share. Two ground measurements agreeing is not two pieces of evidence if both were timed by the same clock.

ASK 10W

Which measurements moved on their own, and which moved together?

BEHIND THE BUTTON 128W

Why agreeing is not the same as independent. Both ground measurements work out where the spacecraft is from how long a signal took. If the clock they both use is wrong, both answers move the same way at the same moment. They will agree beautifully and both be wrong.

What the star camera is doing on the board. It works out where the spacecraft is by looking at stars, with no clock involved. So it either confirms the two ground answers or it does not, and either way it is telling you something they cannot.

Why this matters before anybody acts. Correcting a course that is not wrong makes it wrong, and spends fuel doing it. Finding out which measurements moved on their own is what stops that.

TAKES AS READ

several instruments can be wrong together if they share a part

VERDICT 74W

Agreement counts only when the things agreeing could have failed apart. These measurements all use the same ground clock, so one clock error moves them together. The star camera does not use that clock, and it still matches the prediction. Nothing pushed the spacecraft, and the onboard readings show no push either. Redo the sums with the corrected clock and nearly all the error disappears. Several agreeing numbers turned out to be one fault.

TAKEAWAY 12W

When measurements agree, ask whether they agree separately or share a part.

9.2 How much room is left

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A PERSON · BALLPARK

SCENE 28W

The entry corridor is forty kilometres wide. The circle of doubt around the spacecraft's position is ten kilometres across. Sandoval needs a margin before the burn is called.

GUIDE 58W

Three numbers, and one of them is a single one of the two errors. That was used earlier to work out the circle of doubt. Ask which two this needs. The doubt sits around the middle, so it uses up room on both sides at once. That is why the answer is not the corridor minus the doubt.

ASK 15W

How much clear space is left on each side, if it flies down the middle?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a position is the middle of a circle of doubt, not a fact

VERDICT 75W

The corridor is forty kilometres wide, so aiming down the middle puts twenty either side. The spacecraft could be five kilometres off in any direction, so five of those twenty are already spent before anybody makes a mistake. Fifteen is what is left. That is the number the burn is judged against, and it is why the doubt has to be drawn at the same scale as the corridor rather than mentioned in a footnote.

TAKEAWAY 17W

A route is planned against the doubt, not against the line drawn through the middle of it.

9.3 Say it before you do it

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · VERIFY

SCENE 26W

The correction burn is ready. It adds 3 metres a second, 12 hours before the spacecraft reaches the air. Carter wants the prediction written down first.

GUIDE 54W

Write down how far you expect the burn to move the landing point, before it is fired. Once written it cannot be changed. Then fire the burn, and then decide whether to spend what it costs to measure where the landing point actually went. Not measuring is an answer too, and a poor one.

ASK 14W

Predict how far the burn moves the landing point, fire it, and then measure.

BEHIND THE BUTTON 142W

Why a small burn moves the landing point so much. Three metres a second, twelve hours out, is applied to a spacecraft travelling thousands of metres a second. It barely changes the speed. What it changes is the path, and twelve hours of flying turns a small change of path into a large change of where it arrives.

Why the prediction goes first. A number written after the measurement is not a prediction. With the answer in front of you it is genuinely hard to remember what you expected, which is why the order is enforced rather than suggested.

What is lost by not measuring. A prediction nobody checks can never be wrong, which is what makes it worthless. Checking is also the only way to find out whether the model used to make it can be trusted for the next burn.

TAKES AS READ

a speed and a time give a distance · a prediction is only a test if it is made before the result is known

VERDICT 94W

Three metres every second sounds like nothing. Keep it up for 12 hours and it is a long way. Three metres a second is about 11 kilometres an hour. Eleven, for 12 hours, is roughly 130 kilometres. Say that number out loud first. It is the only thing the tracking pass can be compared against. An engine can run short, point a little wrong, or stop early. None of that shows on the console that told it to fire. The pass afterwards is what turns a burn that happened into a burn that worked.

TAKEAWAY 16W

A burn that happened is not a burn that worked. Only the measurement afterwards says so.

9.4 How a spacecraft turns

ROTATION & STRUCTURES · STRUCTURES TEST BAY · A A PLACE · CALLBACK ·
PROTOCOL

SCENE 27W

The guidance computer is out and the crew is about to fly by hand. Before anybody touches a thruster, the room says plainly what each change does.

GUIDE 61W

Four changes on the left, four results on the right. Pair them by asking what each change does to two things. How hard is it being turned? And how hard is it to turn? One of the four has nothing pushing from outside at all, which is what makes it the odd one. The crew will feel every one of these.

ASK 10W

Match each change to what it does to the turning.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

a push off to one side is what makes something start or stop turning

VERDICT 80W

Four rules cover every hand-flown turn. The same push further from the middle turns harder, which is the one the crew can actually choose. The same push on a heavier, more spread-out craft turns it more slowly, and stops it more slowly too. A push one way and an equal push back cancels out, which is what starting and stopping a turn actually is. And moving things about inside changes nothing overall, because nothing inside can push on the outside.

TAKEAWAY 14W

Turning follows the same rules as moving. It is just harder to see them.

END OF THE SHIFT 23W

Use random and systematic error, weighted averages, errors that move together, propagation, decision thresholds as an evidence chain rather than as isolated facts.

Reentry

BEFORE THE SHIFT — TAG

Catch Lindqvist before the shift changes

Lindqvist, the flight dynamics integrator, has the hand-written power budget from the last pass, and the handover is in six minutes. After that she is off shift for twelve hours. Nothing this morning can be worked out without those figures.

PLAN CARD 77W

A hundred and eighteen hours in. The landing starts in eleven minutes, four of them with no radio, and none of it can be undone. Today you check the claims one at a time and decide whether this spacecraft is ready. The heat shield has been cold for four days. The battery was charged a way nobody planned. Ruiz lined the spacecraft up by hand, against stars. All of it has to hold at the same moment.

BRIEFING 22W

The final minutes require every physical model and procedure to agree: attitude, center of mass, velocity, atmosphere, thermal shield, communications, and recovery.

WHAT YOU DECIDE 13W

Conduct a claim-by-claim readiness review and execute entry with predefined triggers and authority.

10.1 Four minutes of silence

MISSION INTEGRATION • FLIGHT DIRECTOR • A A PLACE • CHOICE

SCENE 30W

For four minutes in the middle of entry there is no radio at all. The room knows it is coming, and Carter still wants somebody to say why it happens.

GUIDE 53W

Four options, and three of them blame equipment or geometry. Ask of each whether it would give exactly four minutes, every time, in the same part of entry. Something breaking would not. Being behind the Moon does not happen here. What happens instead is that the capsule makes the air around itself glow.

ASK 9W

Why does the radio go quiet for four minutes?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

rubbing something hard makes it hot • radio waves travel through air but not through everything

VERDICT 88W

The capsule hits the air at enormous speed, and the air in front of it has nowhere to go. Being squeezed that hard makes it hot, hot enough to glow, and that glowing layer wraps the whole capsule. Radio waves cannot get through it. The silence is not a fault and nobody can fix it from the ground. It ends by itself when the capsule has slowed enough for the air to stop glowing, and everything that might need a decision has to be settled before it starts.

TAKEAWAY 15W

Some silences are physics rather than faults, and nothing on the ground can shorten one.

10.2 The order the silence forces

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · SEQUENCE

SCENE 27W

Everything now depends on eleven minutes, and four of them have no radio at all. Anything that needs the ground has to happen before the silence starts.

GUIDE 50W

Four minutes with no radio is what puts these in order. Ask of each step whether it needs the ground, and whether it could be done after the silence starts. Anything the ground must agree to happens first. Anything compared with a prediction can only happen once entry has begun.

ASK 12W

Put the entry steps in order around the four minutes of silence.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

nothing can be fixed from the ground while the radio is quiet

VERDICT 75W

Four minutes of silence sets the whole order. Anything that needs a decision from the ground has to be settled first, because during the silence there is nobody to ask. Who is flying gets agreed before the silence, not during it. Watching the slowing and the heating can only start once entry has begun, so it comes after. And the recording runs throughout, because this flight is the only test the predictions will ever get.

TAKEAWAY 16W

When part of a plan cannot be changed, everything that might need changing goes before it.

10.3 What the review should fix

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · CHOICE

SCENE 20W

The crew is aboard the recovery ship and the review board meets Monday. Four proposals compete for three funded places.

GUIDE 54W

All four proposals are worth money, and there are three places. Ask of each whether it removes a cause, or helps you cope with one. The same weakness turned up twice on this flight. Several readings on one circuit, and several measurements on one clock. Both times, things agreeing looked stronger than it was.

ASK 10W

Which proposal deserves first place, because it removes the cause?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a fix that removes a cause outlasts a fix that improves the response

VERDICT 71W

The same weakness appeared twice on this flight. Three pressure readings shared one circuit, and later several tracking numbers shared one clock. Both times, agreement looked far stronger than it was, and both times somebody nearly acted on it. Removing those shared parts is the only proposal that stops it happening again. Keeping better records and rehearsing more both help people cope with it, which is a different and smaller thing.

TAKEAWAY 11W

Fixing what caused it outlasts getting better at coping with it.

10.4 How long can the battery last?

POWER & CIRCUITS · POWER & CIRCUITS BAY · A A PLACE · CALLBACK · BALLPARK

SCENE 43W

The fuel cells are gone, and the crew is on the entry batteries days earlier than any plan allowed. The flight doctor, who watches the crew's health from the ground, wants a number of hours before the next decision about switching things off.

GUIDE 57W

Five numbers, and three belong to other questions. The voltage, the brief peak, and the time to the next burn. Ask of each whether it is an amount stored, or a rate of use. An amount divided by a rate leaves a time. And note what the answer assumes: nothing switching on, and every last bit usable.

ASK 6W

How long will the batteries last?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a store is an amount, and a draw is how fast it is being used

VERDICT 82W

There is a store and there is a draw, and dividing one by the other gives hours. Eighteen kilowatt hours at three kilowatts is six hours. It is an ideal figure. It assumes the draw stays flat and that every last bit of the store is usable, and neither is quite true. A cold battery gives less than its label. It is still the number every argument this shift is measured against, so it is worth having early and worth calling rough.

TAKEAWAY 20W

An amount divided by a rate is a time, and that time is what every other decision gets argued against.

END OF THE SHIFT 19W

Use integrated mechanics, thermodynamics, waves, fluids, uncertainty, ethics of command as an evidence chain rather than as isolated facts.

THE CLOSING CARD

Splashdown at 12:07, three good parachutes, and four minutes of blackout that ended with a voice reading its own altitude. The heat shield that had been cold for four days held. The crew were on the recovery deck inside the hour, all three of them walking.

Every claim this room made about that vehicle was written down before entry, and every one of them was still standing afterwards. That is the whole job.

Card text only — no options, boards or answers. 7,832 words of card prose across 40 stops. Generated from themes/bring_them_home_ms by tools/make-cardbook.mjs.